

Introduction to Applied Meta-Analysis

1. What is an Effect Size?

In applied meta-analysis, we often need a standardized way to compare results from different studies. Effect sizes (ES) serve this role — they summarize the magnitude and direction of a treatment effect, independent of the scale of measurement. One common type is the Standardized Mean Difference (SMD).

2. t-score vs. Effect Size

A t-score tells you whether the difference between two groups is statistically significant, taking into account sample size and variance. It reflects how likely the observed difference could occur by chance under the null hypothesis.

An effect size, by contrast, tells you how large that difference actually is in standardized units. While a t-test answers “Is there a difference?”, an effect size answers “How big is the difference?”.

3. Cohen's d

Cohen's d is the most common standardized mean difference (SMD). It is calculated as:

$$d = (M_1 - M_2) / SD_{\text{pooled}}$$

Where:

M_1 = Mean of the treatment group

M_2 = Mean of the control group

$SD_{\text{pooled}} = \sqrt{(SD_1^2 + SD_2^2) / 2}$

Example:

Treatment group: $M_1 = 80$, $SD_1 = 10$ ($n_1 = 25$)

Control group: $M_2 = 70$, $SD_2 = 12$ ($n_2 = 25$)

Step 1. Compute pooled SD:

$$SD_{\text{pooled}} = \sqrt{((10^2 + 12^2)/2)} = \sqrt{((100 + 144)/2)} = \sqrt{122} = 11.05$$

Step 2. Compute d:

$$d = (80 - 70) / 11.05 = 0.905$$

Interpretation: $d = 0.9$ is a large effect (Cohen's benchmarks: 0.2 = small, 0.5 = medium, 0.8 = large).

4. Hedges' g (Small Sample Correction)

Cohen's d slightly overestimates the population effect size when sample sizes are small. Hedges' g applies a correction factor (J) to adjust for bias:

$$g = J \times d, \text{ where } J = 1 - (3 / (4N - 9))$$

Using the same data:

$$N = n_1 + n_2 = 50$$

$$J = 1 - (3 / (4 \times 50 - 9)) = 1 - (3 / 191) = 0.984$$

$$g = 0.984 \times 0.905 = 0.89$$

Interpretation: Hedges' $g \approx 0.89$, only slightly smaller than Cohen's d, but more accurate for small samples.

5. Glass's Δ (Using Control SD Only)

Glass suggested that in some situations, the control group's standard deviation may better represent the baseline variation. Glass's Δ uses the control SD instead of a pooled SD:

$$\Delta = (M_1 - M_2) / SD_{\text{control}}$$

$$\Delta = (80 - 70) / 12 = 0.83$$

Interpretation: Glass's $\Delta = 0.83$, slightly smaller than d or g because the control group had higher variability.

6. Comparison Summary

Estimator	Formula	Use Case	Example Result
-----------	---------	----------	----------------

Cohen's d	$(M_1 - M_2) / SD_{pooled}$	Default SMD	0.91
Hedges' g	$J \times d$	Small sample correction	0.89
Glass's Δ	$(M_1 - M_2) / SD_{control}$	Unequal variances	0.83

Key Takeaway: Effect sizes provide a common language for comparing results across studies. Understanding Cohen's d, Hedges' g, and Glass's Δ is the foundation for meta-analysis.